# Nathaniel Bottman

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Machine learning researcher working on reinforcement-learning post-training and evaluation of large language models: when training teaches a model to report something other than what it has computed, and label-free ways to detect when its reasoning fails. PI of a DARPA-funded research program ("An Operad-Based Framework for Deep Learning," $836K). Previously a research group leader at the Max Planck Institute for Mathematics.

## Experience

**Incubilate** – Founder & Principal Scientist Seattle, WA, July 2025–present

- PI of "An Operad-Based Framework for Deep Learning" ($836K, DARPA AIQ program): operads as the mathematical structure of question decomposition, and label-free grading of multi-step reasoning.
- Showed when verifiable-reward RL teaches a model to bluff (preprint, Nov 2026, with A. Oblomkov): a first-order condition for when a GRPO step makes a model more honest, tested on Datalog (3B), two code models, and abstention on math. No run above the predicted threshold collapses; near it, honesty is a race that a checker can win, and a model of the race predicted held-out runs in advance. A generous-then-cut schedule keeps a model at least 67% honest where training from the start falls to 4–15%.
- Developed *operadic consistency* (OC): tracks accuracy at $r = 0.86$–$0.94$ across twelve LLMs on four multi-hop QA benchmarks, predicts a held-out model's accuracy to 3.0 pp with no labels, and improves selective prediction over a tuned self-consistency baseline at equal cost.
- Extended OC to frontier reasoning models (DeepSeek-R1, Kimi-K2.6, GLM-5.1) by extracting the decomposition from the model's own chain of thought and re-deriving it; positive lift on all 16 selective-prediction cells tested.
- Integrated OC as an accuracy predictor in a DARPA evaluation harness (open-weights 4B–671B and API models).

**Max Planck Institute for Mathematics** – Research Group Leader (W2) Bonn, Jan 2021–Jun 2025

- Led a research group in symplectic geometry; supervised six postdoctoral researchers and multiple student theses; served on the institute's Scientific Committee (hiring).

**Postdoc positions:** U. Southern California (2019–20), Inst. for Advanced Study (2016–19), Northeastern (2015–16).

## Selected Publications

1. **Honesty is a race: predicting when verifiable-reward RL teaches a model to bluff.** N. Bottman, A. Oblomkov. Preprint, 2026. Blog post: `natebottman.com/blog/honesty-is-a-race`.
2. **Operadic consistency: a label-free signal for compositional reasoning failures in LLMs.** N. Bottman, Y. Liu, K. Richardson. arXiv:2606.13649. NeurIPS 2026 (poster). Code on GitHub.
3. **Operads for compositional reasoning in LLMs.** N. Bottman, K. Richardson. arXiv:2606.13634. Presented at the Compositional Learning and the Combining Theory and Benchmarks workshops, ICML 2026.

Twenty further publications in symplectic geometry, combinatorics, and applied mathematics (*Bull. AMS*, *Selecta Math.*, *Algebr. Geom. Topol.*, *J. Symplectic Geom.*), including a polytopal realization of higher-categorical associahedra found with OpenAI's Astra model and formalized in Lean 4 (arXiv:2609.25316).

## Grants & Fellowships

- DARPA AIQ program, "An Operad-Based Framework for Deep Learning," $836K, 2025–present (PI).
- NSF Standard Grant DMS-1906220, $144K, 2019–2022 (PI).
- NSF Mathematical Sciences Postdoctoral Research Fellowship, 2016–2019.

## Education

**Massachusetts Institute of Technology**, Ph.D., Mathematics, 2015. Advisor: Katrin Wehrheim.
**University of Washington**, B.S. Mathematics and B.A. Slavic Languages & Literatures, 2010. Dean's Medal (top graduating senior in the natural sciences).

## Service & Community

Co-organizer, Combining Theory and Benchmarks Workshop, ICML 2026 (Seoul); co-organizer of research seminars and workshops since 2012. 45+ invited talks.

## Technical

Python (GRPO and RL experiments, LLM evaluation pipelines, linear probes), PyTorch.